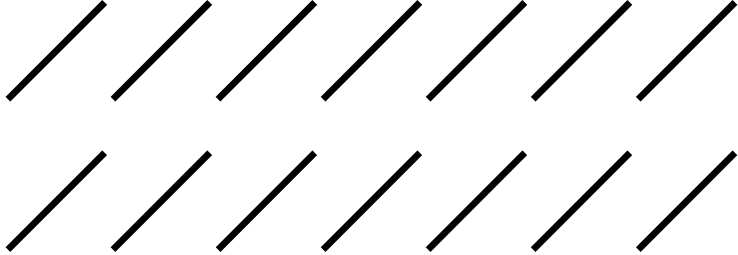


DRAFTING SEQUENCE DIAGRAMS

AGENDA

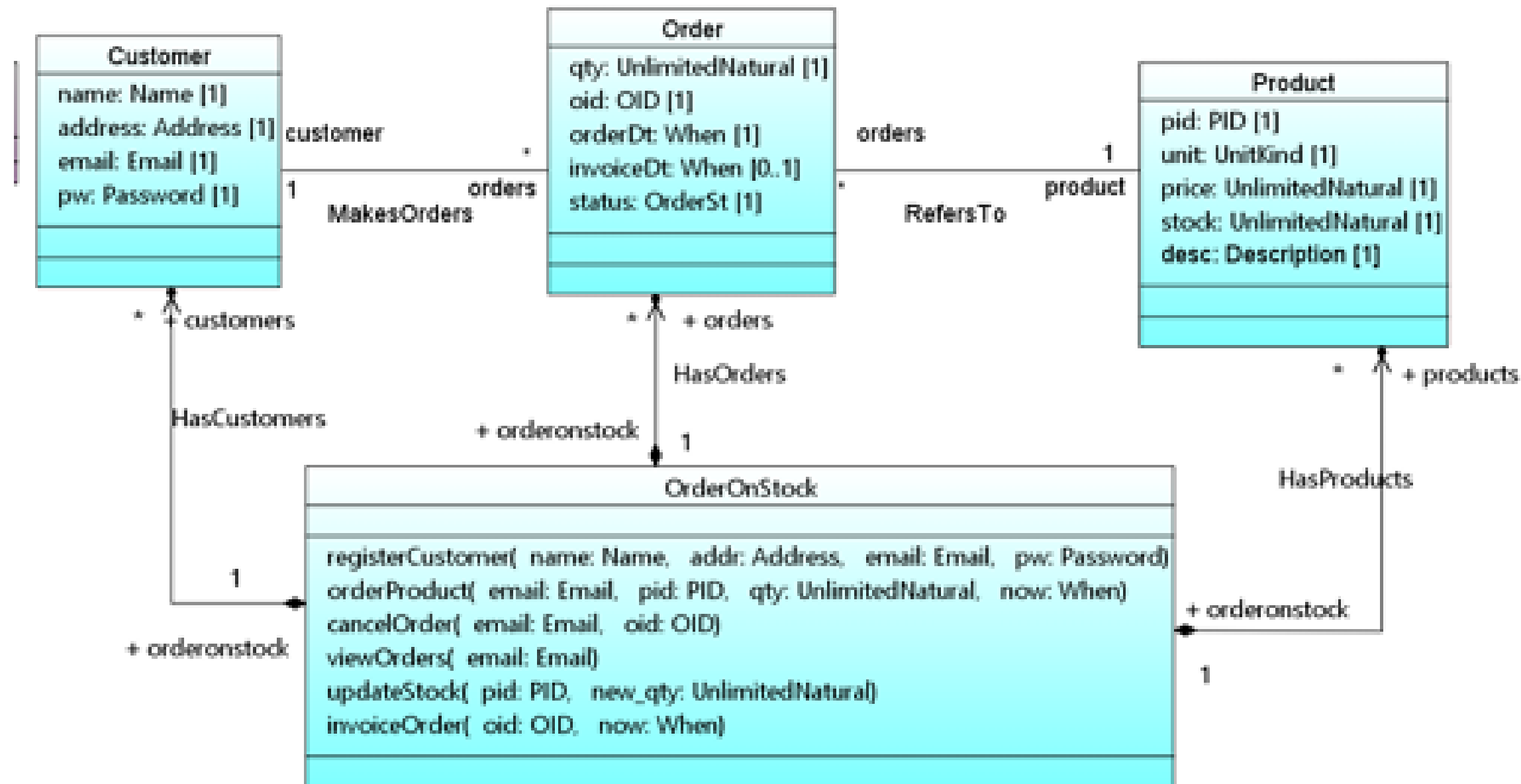
- 
- 01 Simple Practice
 - 02 Identifying classes
 - 03 Identifying methods and associations
 - 04 Drafting sequence diagram
 - 05 Identifying alt/opt/loop envelopes
 - 06 Linking to Next Weeks

section 1

SIMPLE PRACTICE

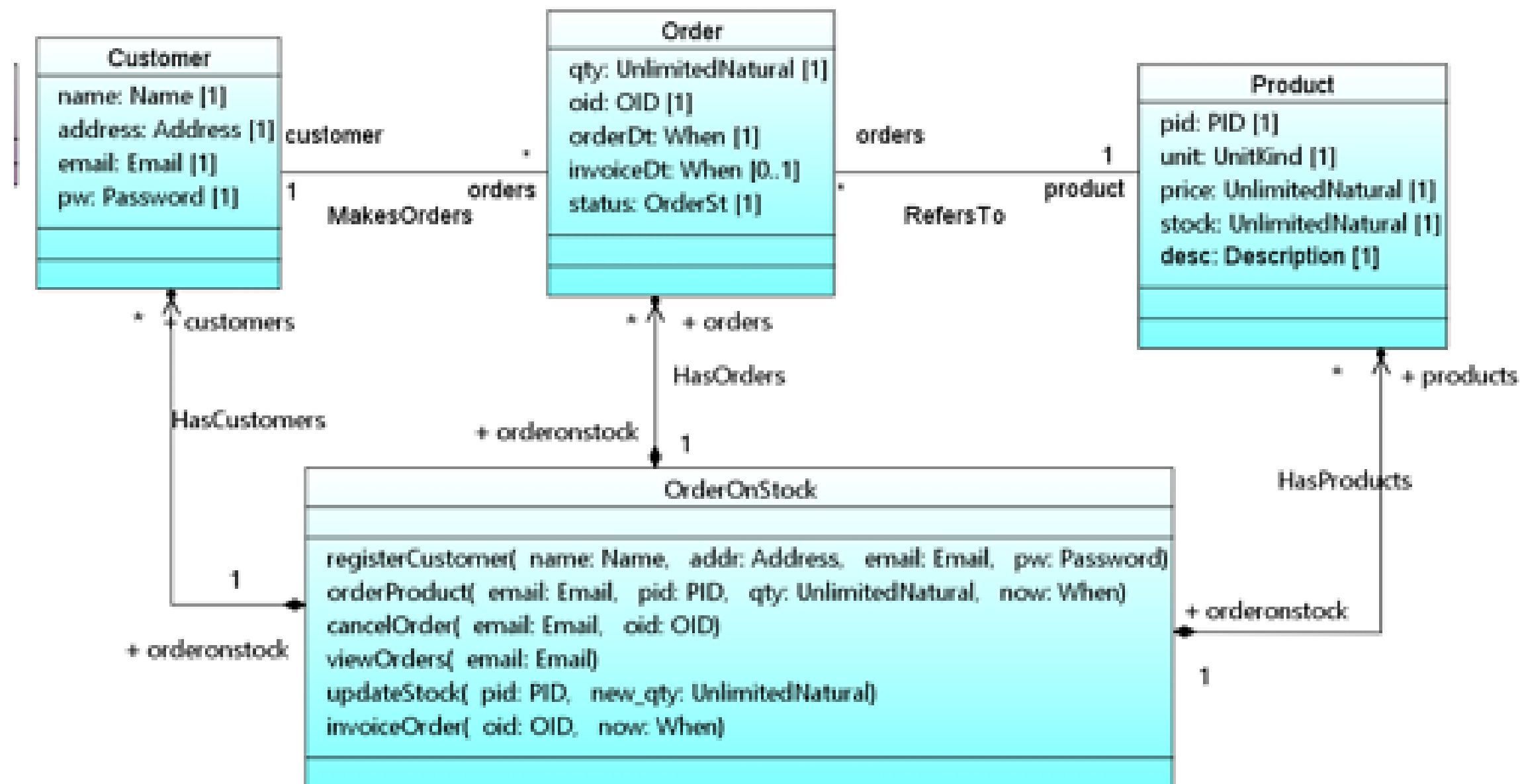
Name	Customer orders product
Purpose	Customer orders a product and expects the order process to succeed if they and the product exist.
Prerequisites	N/A
Narrative	1. The customer requests to order a product by providing their email, the product ID, and the desired quantity. 2. The system checks if the customer exists by searching for the provided email. 3. The system checks if the product exists by searching for the provided product ID. 4. The system creates a new order using the customer, product, and quantity information. 5. The system stores the order and returns the order ID to the caller.
Outcome	• A new order is created and stored in the system. • The order ID is returned to the customer.
Alternative Narrative	The system does not find the product requested by the customer.
Alternative Narrative Outcome	The system displays an error message to the customer.
Alternative Narrative	The system does not find a customer with the provided email address.
Alternative Narrative Outcome	The system displays an error message to the customer.

1.1. Drafting Sequence Diagrams - Simple Practice



Draft a sequence diagram for the Customer Orders Product use case specification, which uses relevant classes and methods from the above class diagram.

1.1. Drafting Sequence Diagrams - Simple Practice



HasCustomers = there is a List<Customer> in the OrderOnStock class

HasOrders = there is a List<Order> in the OrderOnStock class

HasProducts = there is a List<Product> in the OrderOnStock class

section 2

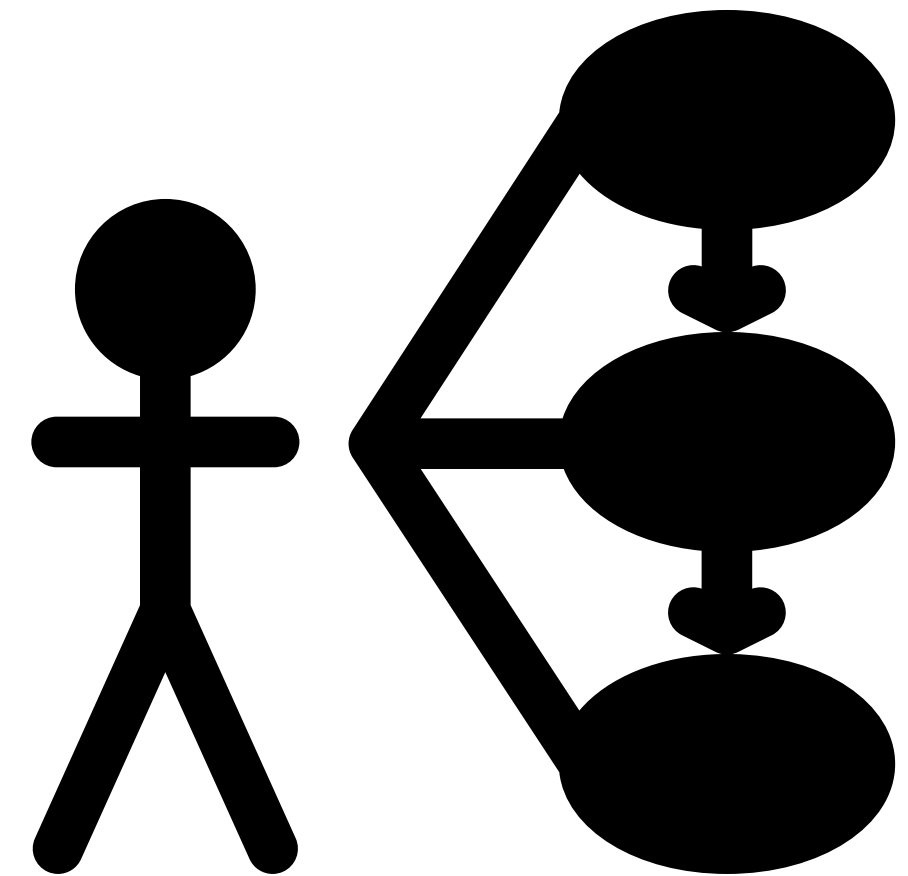
IDENTIFYING CLASSES

2.1. Identifying classes

Given your use case specifications, the nouns in the main narrative and outcome show which classes roughly will be part of the sequence diagram.

Activity

- For your use case specification from D1:
 - Which classes are needed to fulfil the narrative and outcome?
 - If there are needed classes missing in the class diagram, revise your draft class diagram.



section 3

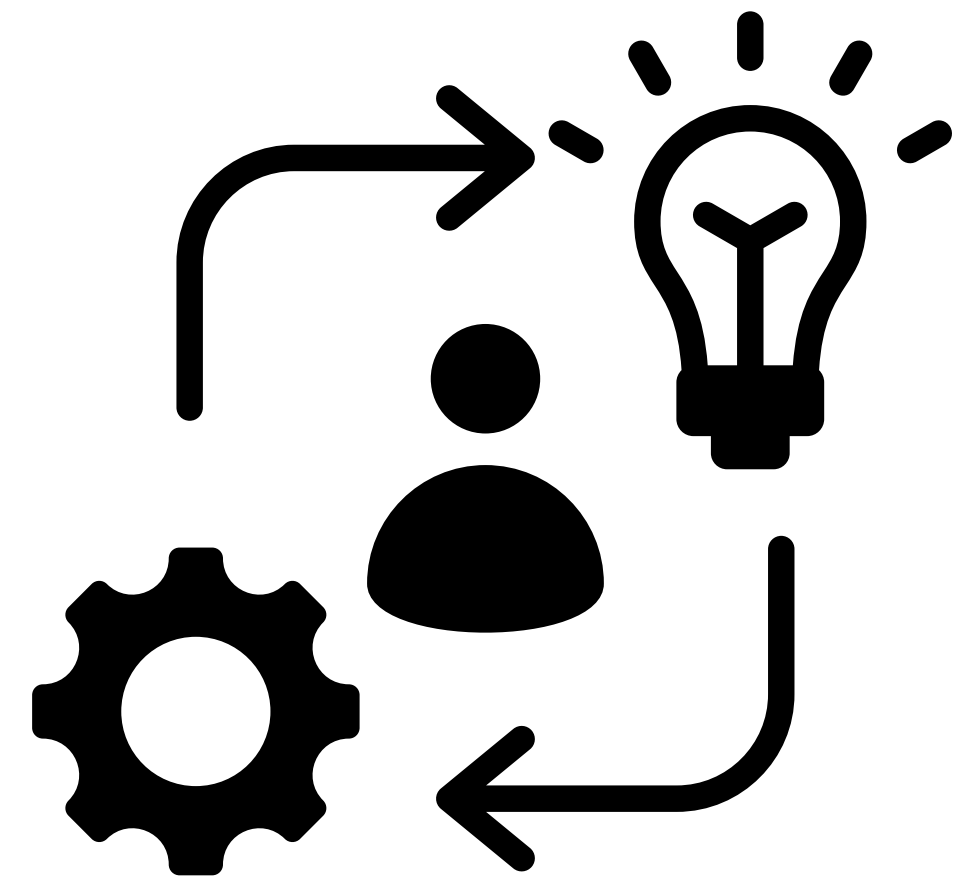
IDENTIFYING METHODS AND ASSOCIATIONS

3.1. Identifying methods and associations

Mapping the main narrative to methods existing in the class diagram is a good start for what should go on the arrows of the sequence diagram.

Activity

- For your use case spec and class diagram:
 - Which methods and associations are needed to fulfil the narrative and outcome?
 - If there are needed methods and associations missing in the class diagram, revise class diagram.



section 4

DRAFTING SEQUENCE DIAGRAM

4.1. Drafting sequence diagram

Once the classes and methods are identified, they can be put into a sequence diagram that reads from top to bottom following the main narrative of the use case.

Activity

- Given the identified classes and methods, draft the sequence diagram including:
 - Objects
 - Life lines
 - Labelled method/return arrows
 - Main trigger



section 5

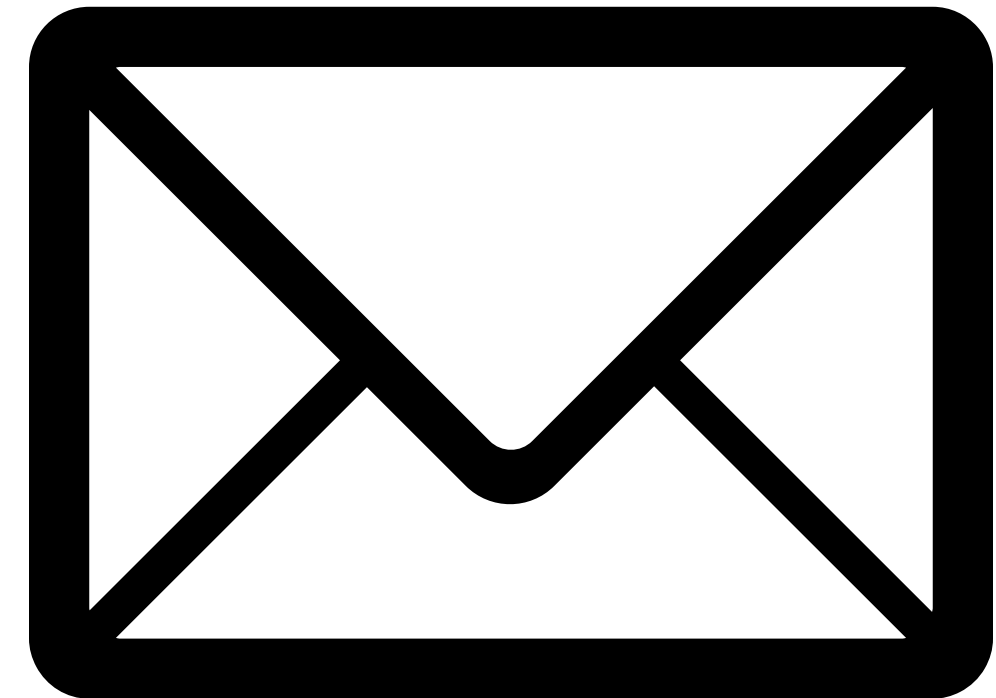
IDENTIFYING ENVELOPES

5.1. Identifying envelopes

Looking at the alternative narratives and the main narrative of the use case specification, you can identify where repetition, optional, or conditional cases will be included in the sequence diagram.

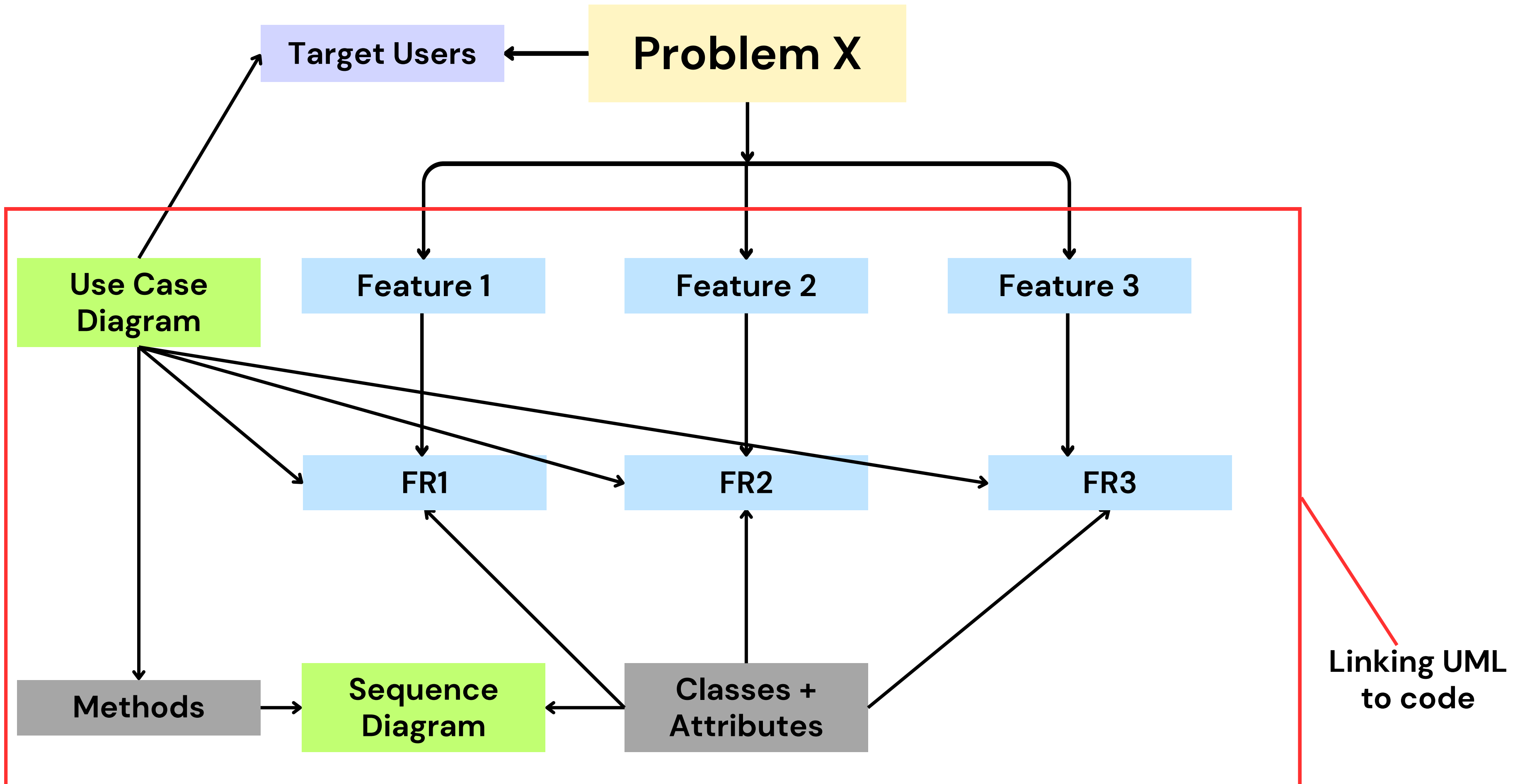
Activity

- Given the use case specification, update the sequence diagram with alt/opt/loop envelopes.
- Ensure that the class diagram includes all the methods present in the sequence diagram.



section 6

LINKING TO NEXT WEEKS



Thank You!

ANONYMOUS FEEDBACK

